PROJECT REPORT SPECTRAL DECOMPOSITION AND ANALYSIS OF COMPLEX SIGNALS

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Subject: Signals and systems

1. Abstract

This project presents the spectral decomposition and analysis of a complex signal using MATLAB. A complex signal was constructed by combining three complex exponential components with frequencies of 10 Hz, 20 Hz and 30 Hz and amplitudes of 10, 5 and 20 respectively. The signal was analyzed in the time domain and frequency domain. Its real part, imaginary part, magnitude and phase were obtained. Fast Fourier Transform (FFT) was then applied to obtain the magnitude and phase spectra. The FFT spectrum was also shifted to represent negative and positive frequencies. Power Spectral Density (PSD) was calculated using Welch's method. Finally, dominant frequency components were automatically detected using the MATLAB findpeaks function. The detected frequencies were approximately 9.995 Hz, 19.990 Hz and 29.985 Hz, which closely match the designed 10 Hz, 20 Hz and 30 Hz components.

2. Introduction

Signals can be represented in both the time domain and the frequency domain. While a time-domain plot shows how a signal varies with time, frequency-domain analysis reveals the frequency components contained in the signal. Spectral analysis is therefore important in communication systems, signal processing, radar, audio processing and many other electronic applications.

In this project, a complex signal containing three known frequency components is generated and analyzed using MATLAB. FFT is used to transform the signal from the time domain to the frequency domain. The magnitude spectrum, shifted spectrum, phase spectrum and PSD are studied, and dominant frequencies are identified automatically.

3. Objectives

• To generate a complex signal containing multiple frequency components. • To observe the real, imaginary, magnitude and phase components of the signal. • To perform FFT-based frequency-domain analysis. • To study positive and negative frequency components using fftshift. • To obtain the phase spectrum of the complex signal. • To calculate the Power Spectral Density using Welch's method. • To automatically identify dominant frequency components using findpeaks. • To compare the detected frequencies with the original designed frequencies.

4. Software and Tools

• MATLAB R2024b • MATLAB functions: fft, fftshift, pwelch, findpeaks, plot and stem • Sampling frequency: 1000 Hz

5. Signal Parameters

Component Frequency Amplitude X1 10 Hz 10 X2 20 Hz 5 X3 30 Hz 20

The sampling frequency is fs = 1000 Hz. The sampling interval is Ts = 1/fs, and the time vector used in the MATLAB program is t = -1:Ts:1.

6. Mathematical Model

The three complex exponential components are defined as:

$X1(t) = 10 \exp(j2\pi(10)t)$

$X2(t) = 5 \exp(j2\pi(20)t)$

$X3(t) = 20 \exp(j2\pi(30)t)$

The complete signal is:

x(t) = X1(t) + X2(t) + X3(t)

7. Methodology

1. Signal generation: Three complex exponentials are generated using $\exp(j2\pi Ft)$ and added to form the **combined complex signal.**

2. Time-domain analysis: The real part, imaginary part, magnitude and phase are calculated using real(), imag(), abs() and angle().

3. FFT analysis: The FFT of the complex signal is calculated using fft(). The magnitude is normalized by the number of samples.

4. Frequency-axis construction: The frequency resolution is calculated as df = fs/n and a corresponding frequency vector is generated.

5. Shifted spectrum: fftshift() is used to place the zero-frequency component at the center so that negative and positive frequencies can be viewed together.

6. Phase spectrum: The phase of the shifted FFT is obtained using angle().

7. PSD analysis: Welch's method is used through pwelch() to estimate the power spectral density.

8. Dominant-frequency detection: findpeaks() is applied to the normalized FFT magnitude. A minimum peak height of 0.5 and minimum peak distance of 5 Hz are used to avoid very small or closely spaced unwanted peaks.

8. MATLAB Program

```
clc; close all; clear;
%construction % Parameters
F1= 10; F2=20; F3=30;
A1= 10; A2=5; A3=20; fs = 1000;
Ts = 1/fs; t = -1:Ts:1;
% Signals generation X1=A1*exp(1i*2*pi*F1*t); X2=A2*exp(1i*2*pi*F2*t);
X3=A3*exp(1i*2*pi*F3*t);
x=X1 + X2 +X3;
```

```
% Signal components
 x_real = real(x);
 x_imaginary = imag(x);
x_mag = abs(x);
X_phase = angle(x);
% Plotting figure;
subplot(3,3,1) plot(t,x_real); xlabel('Time (s)');
 ylabel('Amplitude');
title('Real Part');
 grid on;
subplot(3,3,2) plot(t,x_imaginary);
xlabel('Time (s)');
ylabel('Amplitude');
title('Imaginary Part');
grid on;
subplot(3,3,3) plot(t,x_mag);
xlabel('Time (s)');
ylabel('Magnitude');
title('Magnitude');
grid on;
subplot(3,3,4) plot(t,X_phase);
xlabel('Time (s)');
ylabel('Phase (rad)');
title('Phase');
grid on;
%applying fft n = length(x);
df = fs/n;
f = (0:n-1)*df;
y = fft(x);
y_mag=abs(y)/n;
```

```
subplot(3,3,5) stem(f,y_mag);
xlabel('Frequency (Hz)');
ylabel('Magnitude');
 title('FFT Magnitude Spectrum');
grid on; xlim([0 35]);
% shifting to -ve y_shift=fftshift(y);
y_shift_mag=abs(y_shift)/n;
f_shift = (-floor(n/2):
ceil(n/2)-1)*df;
subplot(3,3,6) stem(f_shift,y_shift_mag);
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('FFT Magnitude Spectrum');
grid on;
xlim([-35 35]);
%to find phase of shifted
fft y_shift_phase=angle(y_shift);
subplot(3,3,7) stem(f_shift, y_shift_phase);
xlabel('Frequency (Hz)');
ylabel('Phase (rad)');
title('FFT Phase Spectrum'); grid on;
 xlim([-35 35]);
% Calculate PSD using Welch method
[pxx, f_psd] = pwelch(x, [], [], [], fs, 'centered');
subplot(3,3,8) plot(f_psd, pxx);
 grid on;
xlabel('Frequency (Hz)');
 ylabel('Power/Frequency (w/Hz)');
 title('Power Spectral Density of Signal');
xlim([-35 35]);
```

```
% Find dominant frequency components
[peaks, locations] = findpeaks(y_mag, f, ... 'MinPeakHeight', 0.5, ... 'MinPeakDistance', 5);
% Display dominant frequencies disp('Dominant Frequencies (Hz):');
disp(locations);
% Display magnitude of dominant frequencies
 disp('Magnitude of Dominant Frequencies:');
 disp(peaks);
% Plot dominant frequencies
subplot(3,3,9) stem(f, y_mag);
 hold on;
plot(locations, peaks, 'ro', 'MarkerSize', 8);
 xlabel('Frequency (Hz)');
 ylabel('Magnitude');
title('Dominant Frequency Components');
grid on; xlim([0 35]);
hold off;
```

9. Results

Expected Component Detected Frequency Detected Magnitude 10 Hz 9.9950 Hz 10.0183 20 Hz 19.9900 Hz 5.0216 30 Hz 29.9850 Hz 19.9629 The detected frequencies are very close to the original 10 Hz, 20 Hz and 30 Hz values. The small differences are due to the finite sampling interval and FFT frequency-bin resolution. The detected magnitudes are also close to the specified amplitudes of 10, 5 and 20.

The 30 Hz component is the strongest of the three components because its designed amplitude is 20. The 10 Hz and 20 Hz components are also clearly identified by the peak-detection method.

10. Discussion **The time-domain plots show that the combined signal is a complex waveform formed from multiple sinusoidal components. The real and imaginary parts provide two views of the complex signal, while magnitude and phase describe its amplitude and angular information.**

The FFT magnitude spectrum reveals the frequency content of the signal. The use of fftshift allows the spectrum to be viewed around zero frequency, which is useful for understanding negative and positive frequency components of a complex signal.

The PSD obtained using Welch's method provides a power-versus-frequency representation. Dominant- frequency detection using findpeaks makes the analysis automatic instead of relying only on visual inspection.

11. Conclusion

The project successfully demonstrates the spectral decomposition and analysis of a multi-component complex signal using MATLAB. A signal containing 10 Hz, 20 Hz and 30 Hz components was generated and analyzed in both time and frequency domains. FFT, fftshift, phase analysis and Welch PSD estimation were implemented. The dominant-frequency detection correctly identified approximately 9.995 Hz, 19.990 Hz and 29.985 Hz, confirming the presence of the designed frequency components. The project demonstrates how frequency- domain methods can be used to identify and study the individual spectral components of a complex signal.

12. Future Scope

• Extract individual frequency components from the FFT spectrum. • Reconstruct the signal using inverse FFT (IFFT) and compare it with the original signal. • Study the effect of sampling frequency and FFT size on frequency resolution. • Investigate spectral leakage and methods for reducing it. • Add noise to the signal and study its effect on spectral analysis. • Compare FFT-based analysis with other spectral-analysis techniques.

13. References

1. MATLAB R2024b documentation and built-in functions: fft, fftshift, pwelch and findpeaks.

2. Project MATLAB program developed and tested by the student.